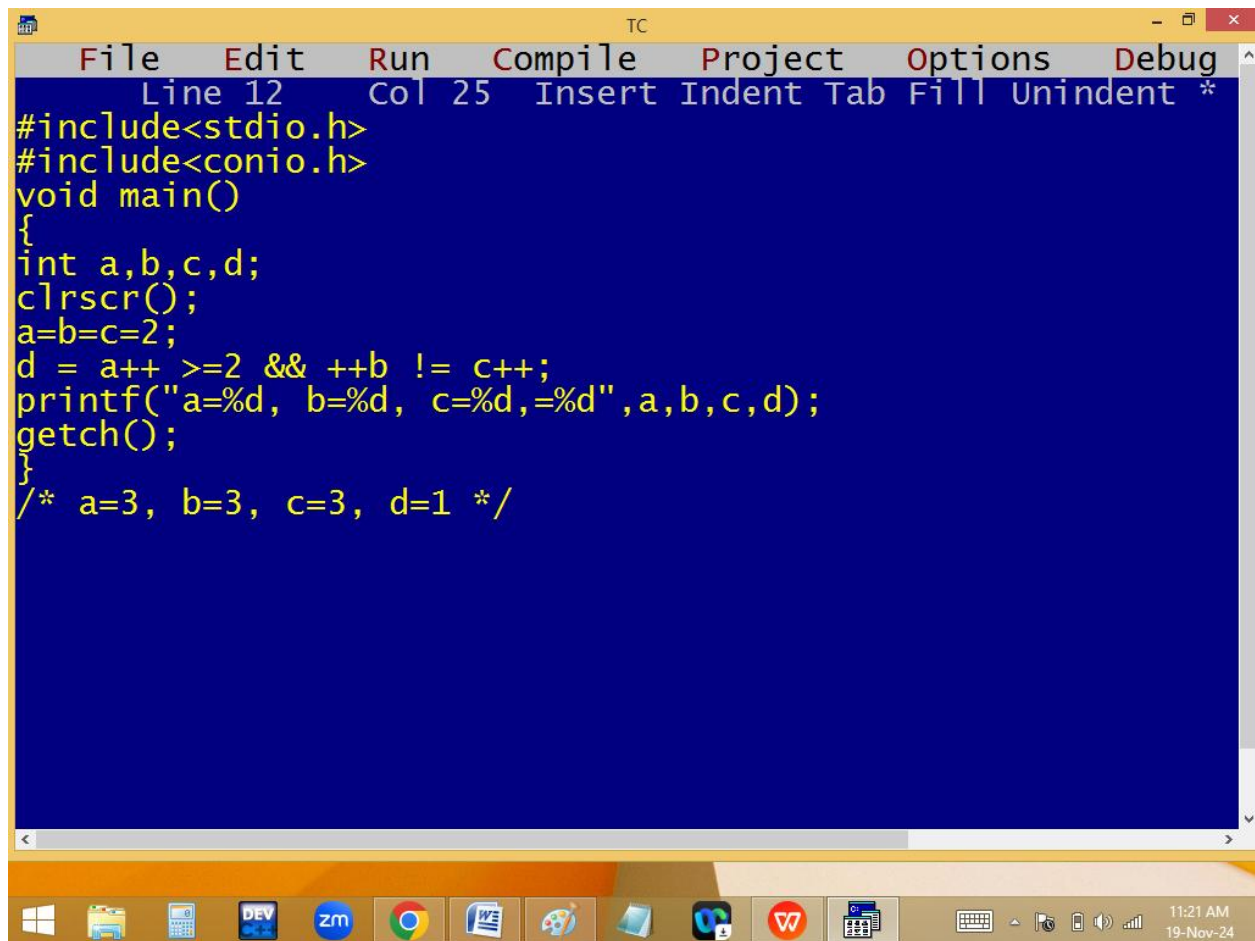




```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 25 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c,d;
clrscr();
a=b=c=2;
d = a++ >=2 && ++b != c++;
printf("a=%d, b=%d, c=%d,=%d",a,b,c,d);
getch();
}
/* a=3, b=3, c=3, d=1 */

11:21 AM
19-Nov-24
```

a=2 b=2 c=2

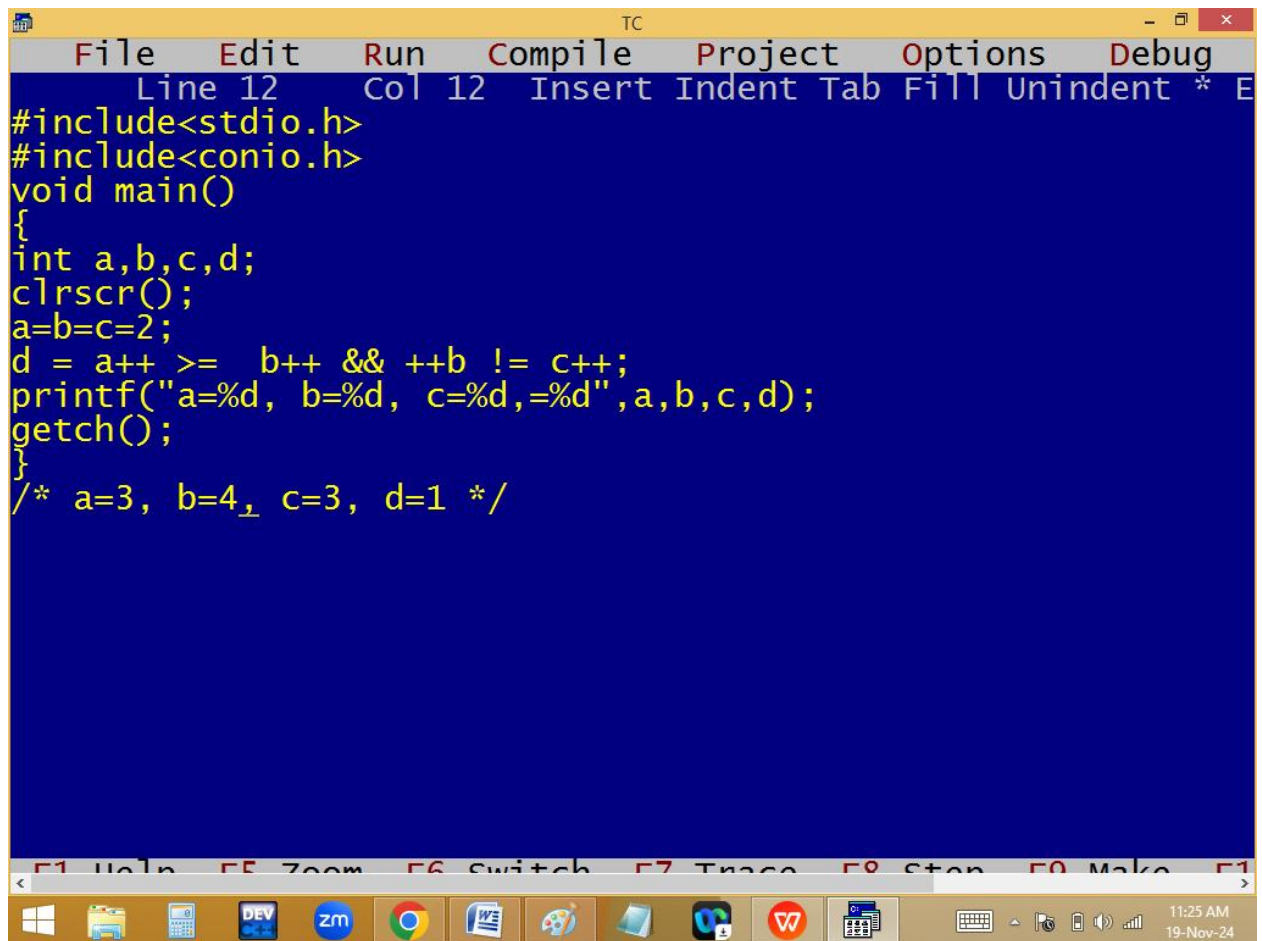
d = a++ >= 2 && ++b != c++;

2 >= 2 3 != 2

1 && 1

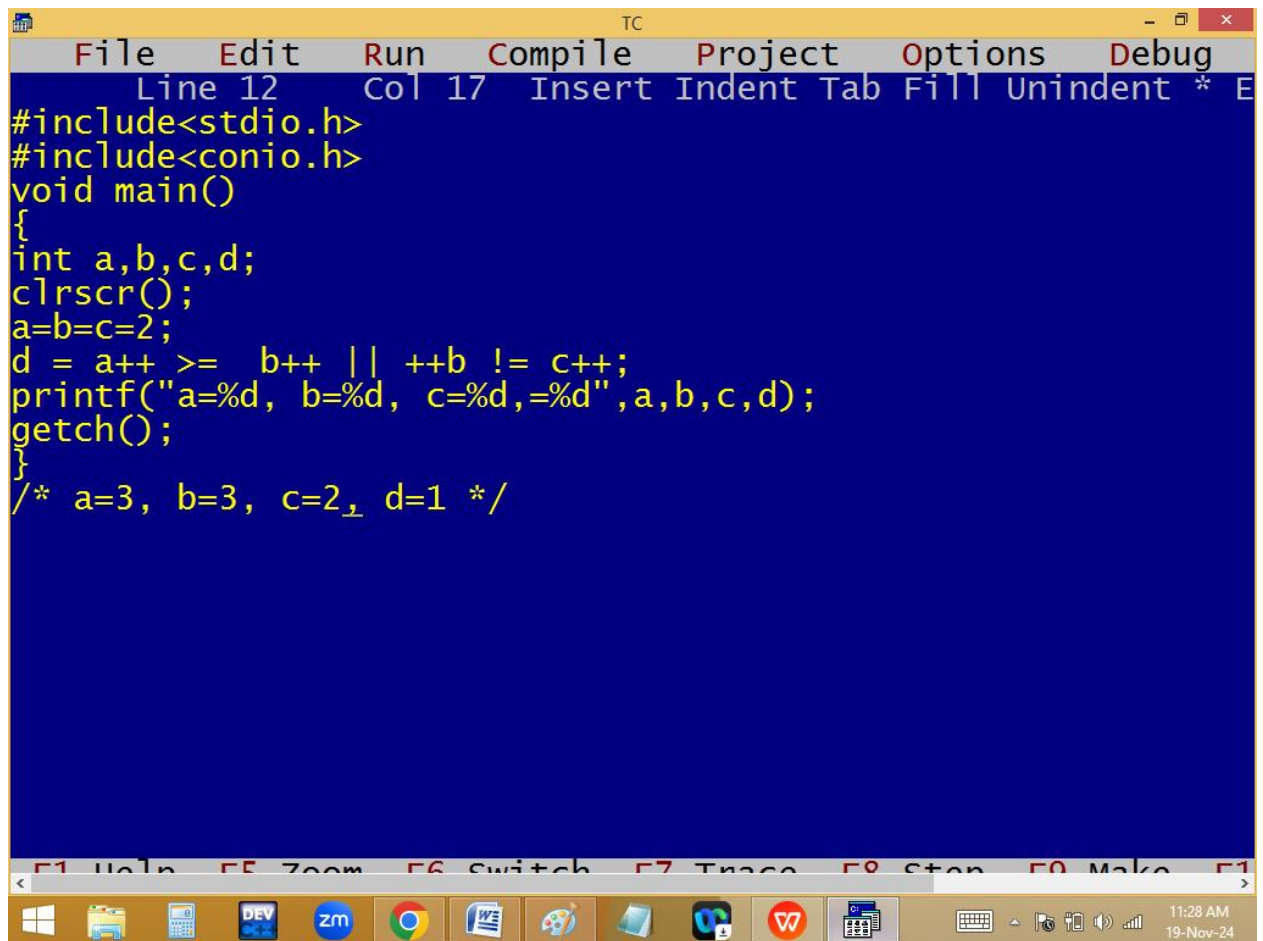
|

The diagram illustrates the evaluation of the expression `d = a++ >= 2 && ++b != c++;` with initial values `a=2, b=2, c=2`.
1. The expression is broken down into sub-expressions: `a++`, `>= 2`, `&&`, `++b`, `!=`, and `c++`.
2. The values of these sub-expressions are calculated: `a++` evaluates to 2, `>= 2` evaluates to 1 (true), `&&` is the logical AND operator, `++b` evaluates to 3, `!=` is the logical NOT EQUAL operator, and `c++` evaluates to 2.
3. The results of the comparisons are shown: `2 >= 2` is 1, and `3 != 2` is 1.
4. The logical AND operation is performed: `1 && 1` results in 1.
5. The final result, 1, is assigned to the variable `d`, as indicated by the curved arrow.



```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 12 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c,d;
clrscr();
a=b=c=2;
d = a++ >= b++ && ++b != c++;
printf("a=%d, b=%d, c=%d,= %d",a,b,c,d);
getch();
}
/* a=3, b=4, c=3, d=1 */

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```



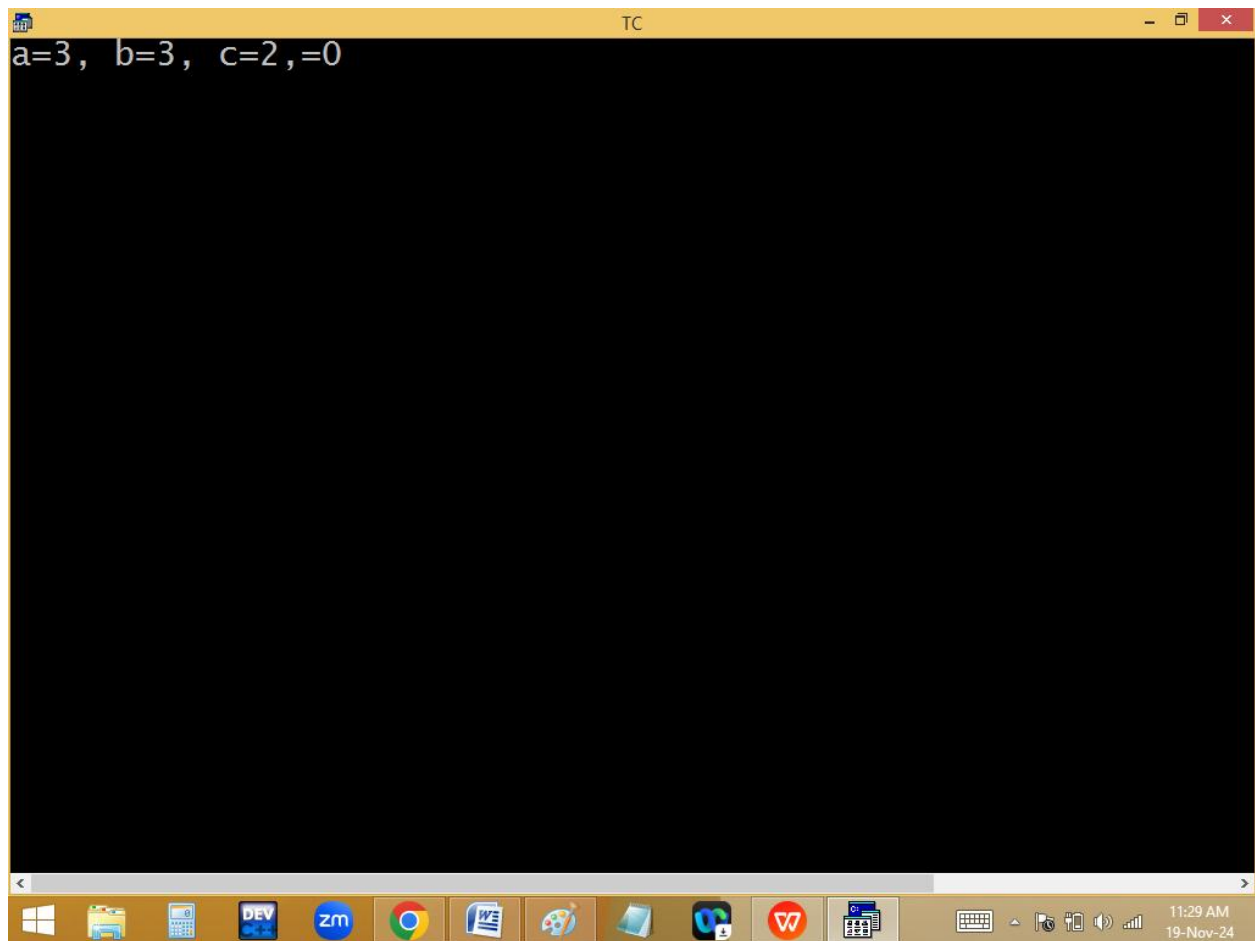
```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 17 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c,d;
clrscr();
a=b=c=2;
d = a++ >= b++ || ++b != c++;
printf("a=%d, b=%d, c=%d,= %d",a,b,c,d);
getch();
}
/* a=3, b=3, c=2, d=1 */

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

a=2 b=2 c=2 ✓

d = a++ >= b++ || ++b != c++;
2 >= 2
Not checked

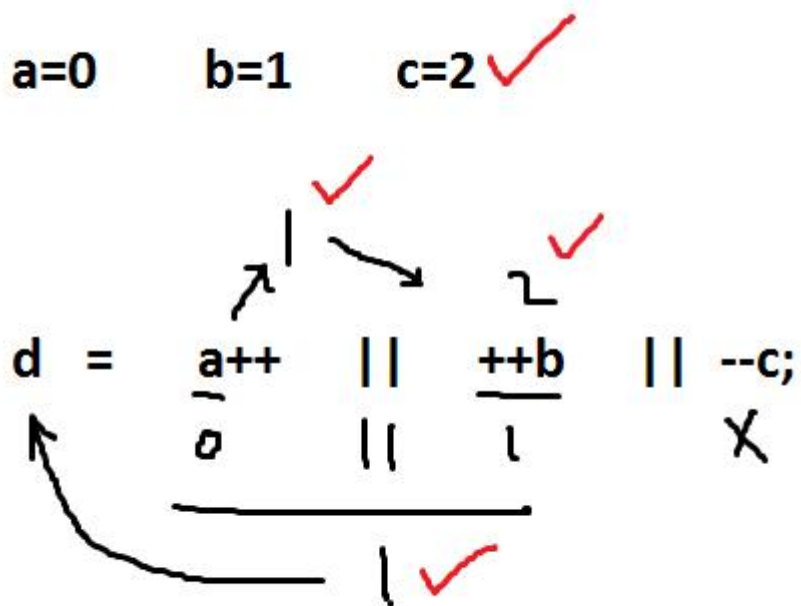
```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 22 Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c,d;
clrscr();
a=b=c=2;
d = a++ != b++ && ++b != c++;
printf("a=%d, b=%d, c=%d, d=%d",a,b,c,d);
getch();
}
/* a=3, b=3, c=2, d=0 */
```



a=2 b=2 c=2 ✓

d = a++ != b++ && ++b != c++;
2 2
Not checked


```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 22 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0, b=1,c=2,d;
clrscr();
d = a++ || ++b || --c;
printf("a=%d, b=%d, c=%d,=%d",a,b,c,d);
getch();
}
/* a=1, b=2, c=2, d=1_*/
```




```

File Edit Run Compile Project Options Debug
Line 11 Col 17 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0, b=1,c=2,d;
clrscr();
d = a++ && ++b || --c;
printf("a=%d, b=%d, c=%d, d=%d",a,b,c,d);
getch();
}
/* a=1, b=1, c=1, d=1 */

```

a=0 b=1 ✓ c=2

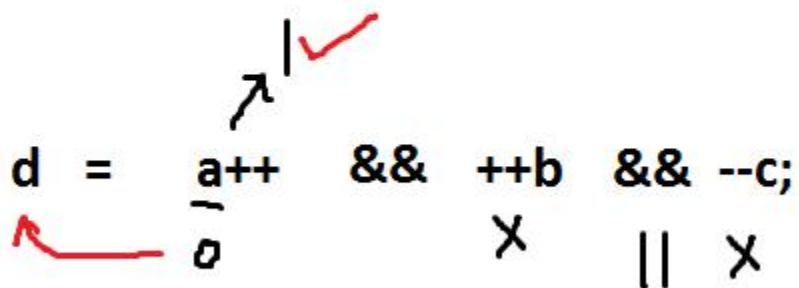
d = a++ && ++b || --c;

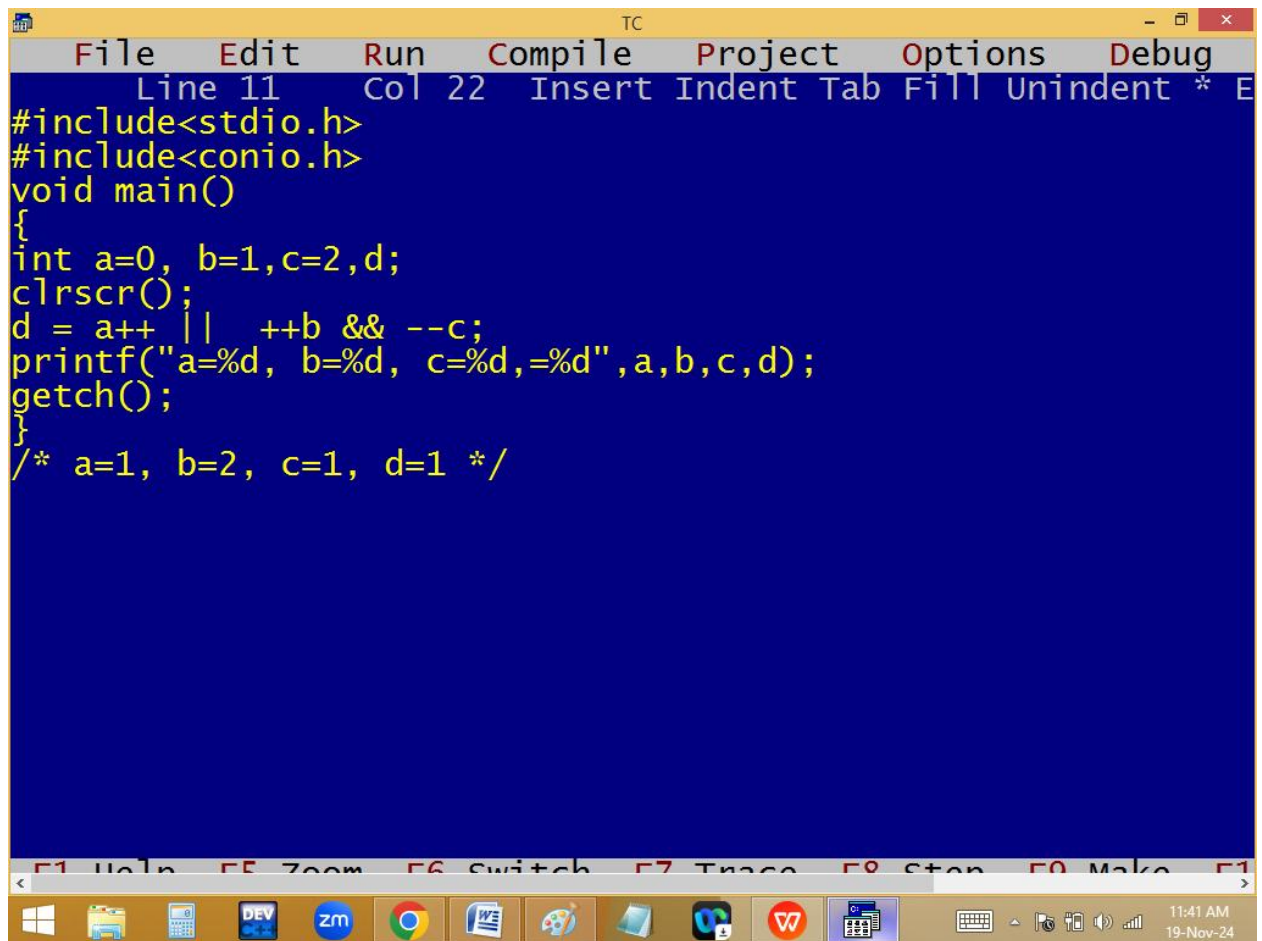
```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 22 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0, b=1,c=2,d;
clrscr();
d = a++ && ++b && --c;
printf("a=%d, b=%d, c=%d, d=%d",a,b,c,d);
getch();
}
/* a=1, b=1, c=2, d=0 */
```



a=0 b=1 ✓ c=2 ✓

d = a++ && ++b && --c;
0 X || X





```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 22 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0, b=1,c=2,d;
clrscr();
d = a++ || ++b && --c;
printf("a=%d, b=%d, c=%d,=%d",a,b,c,d);
getch();
}
/* a=1, b=2, c=1, d=1 */

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

11:41 AM
19-Nov-24

a=0 b=1 ✓ c=2 ✓

d = a++ || ++b && --c;

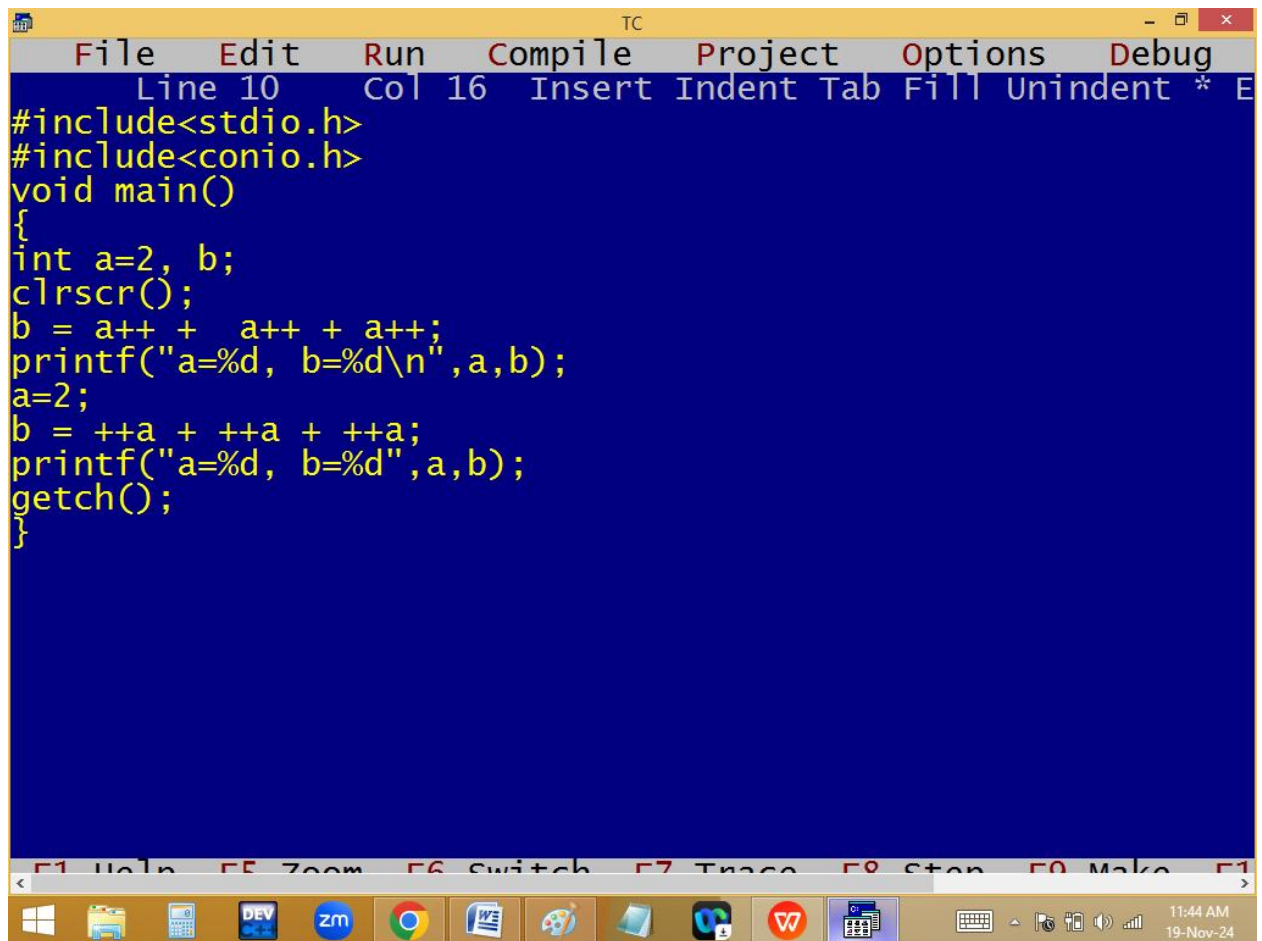
↑ ✓ ↓ 1 ✓ ↑ 1 ✓

0 || 1

 | && |

 | ✓

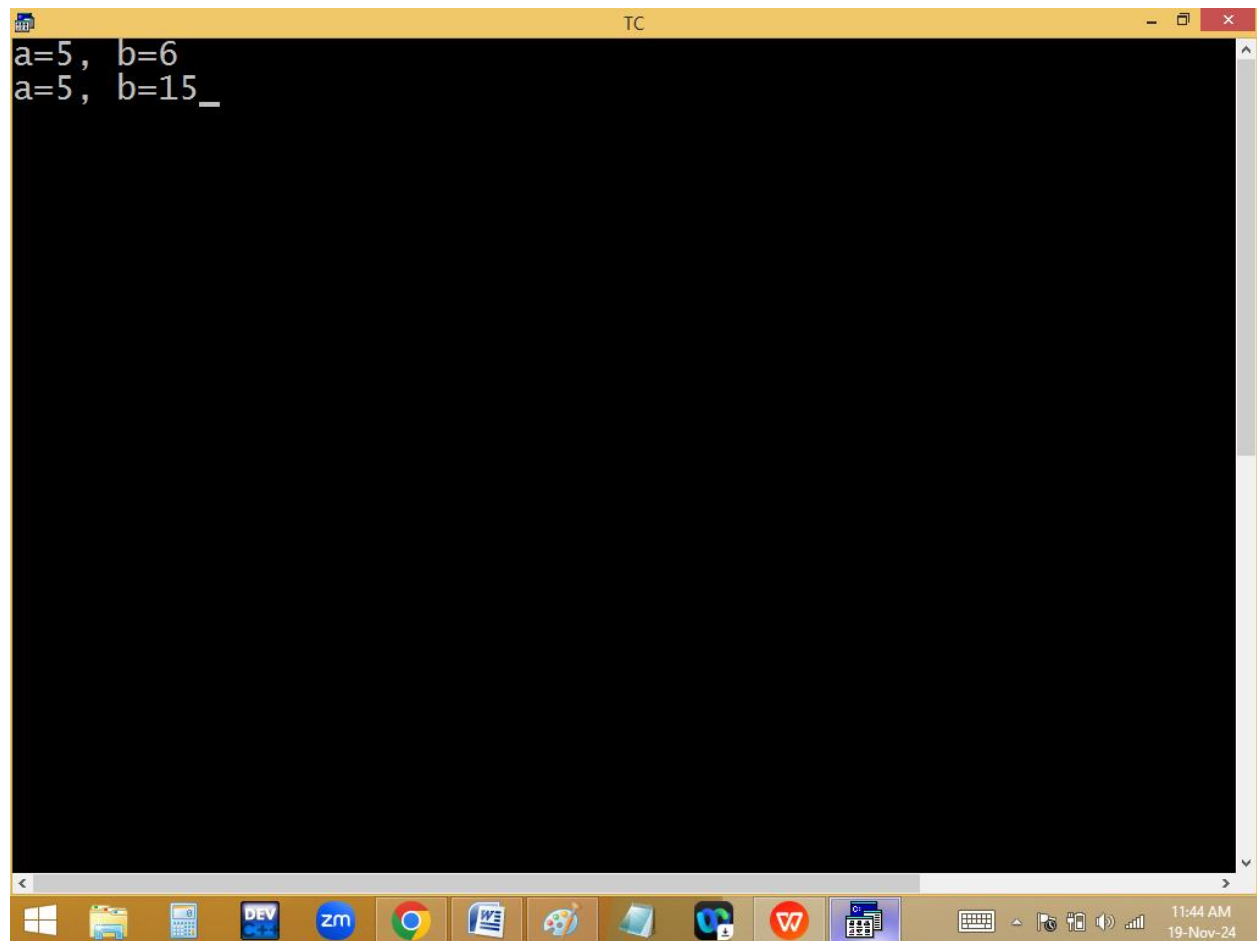
↑

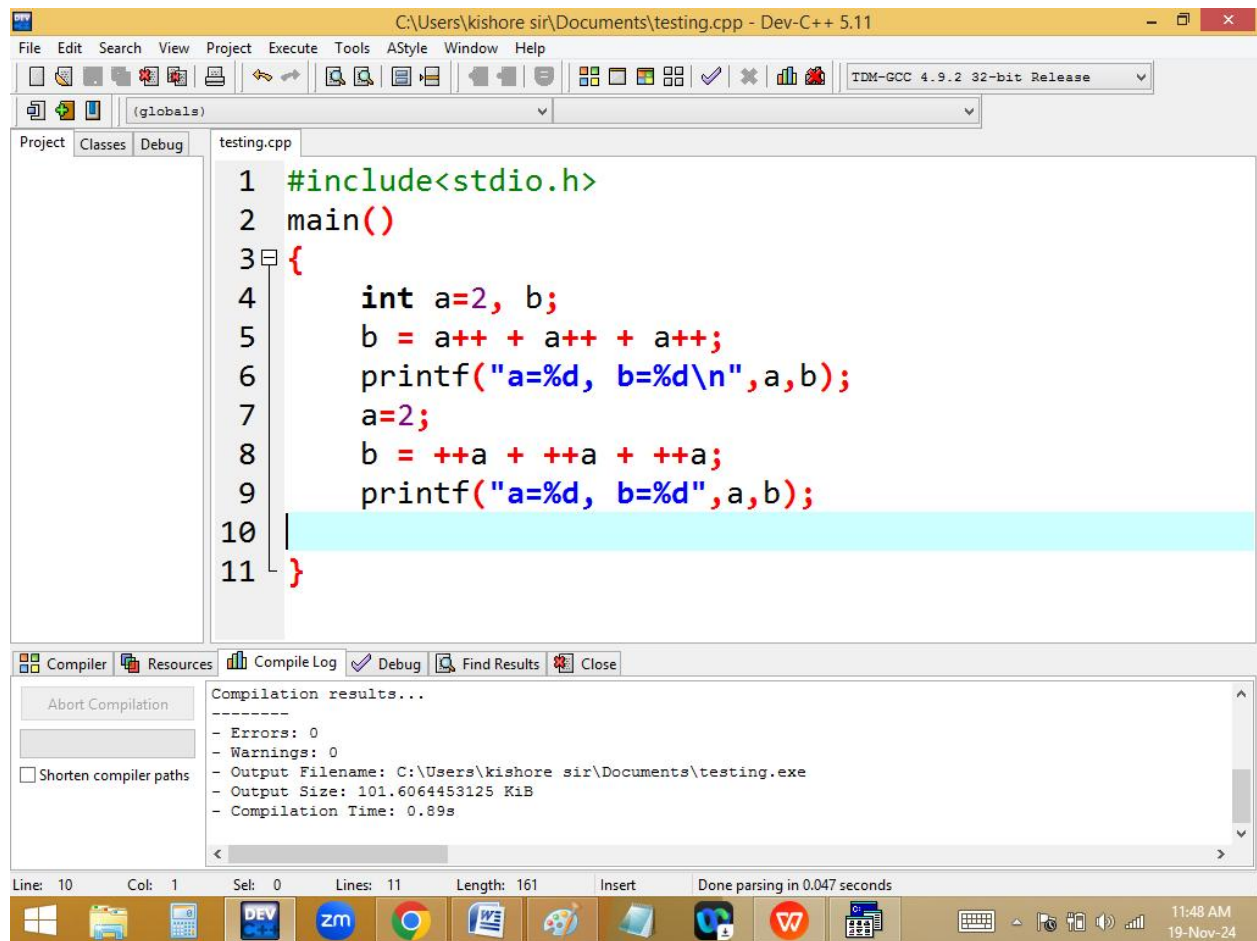


```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 16 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=2, b;
clrscr();
b = a++ + a++ + a++;
printf("a=%d, b=%d\n",a,b);
a=2;
b = ++a + ++a + ++a;
printf("a=%d, b=%d",a,b);
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

11:44 AM
19-Nov-24





```
C:\Users\kishore sir\Documents\testing.exe
a=5, b=9
a=5, b=13
-----
Process exited after 0.01034 seconds with return value 0
Press any key to continue . . .
```

a=2

$b = a++ + a++ + a++;$
 $\uparrow \quad \uparrow \quad \uparrow$
 $\underline{2} + \underline{3} + \underline{4} = 9 \checkmark$

a=2 3 4 5 ✓

$b = ++a + ++a + ++a;$
 $\uparrow \quad \uparrow \quad \uparrow$
 $\underline{4} + \underline{4}$
 $\underline{8} + 5 = 13$

Compound assignment / short hand operators: Here we are using one assignment operator with the combination other operators as follows.

$+=$, $-=$, $*=$, $\%=$, $/=$, $<<=$, $>>=$, $\sim=$,

Eg:

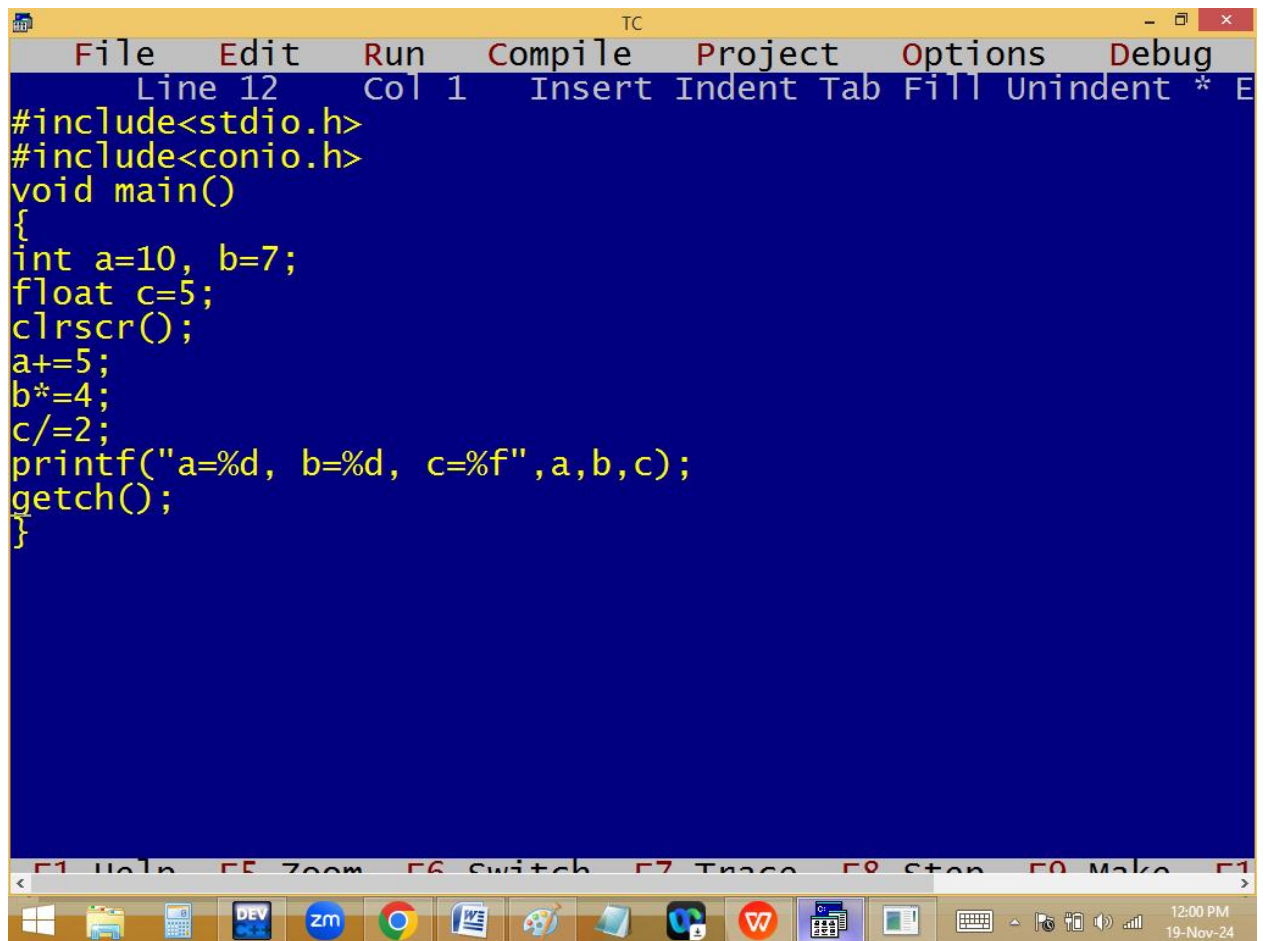
Int a=10, b=7;

Float c=5;

a+=5; i.e. $a=a+5 \rightarrow a=10+5=15$

b*=4; i.e. $b=b*4 \rightarrow b=7*4=28$

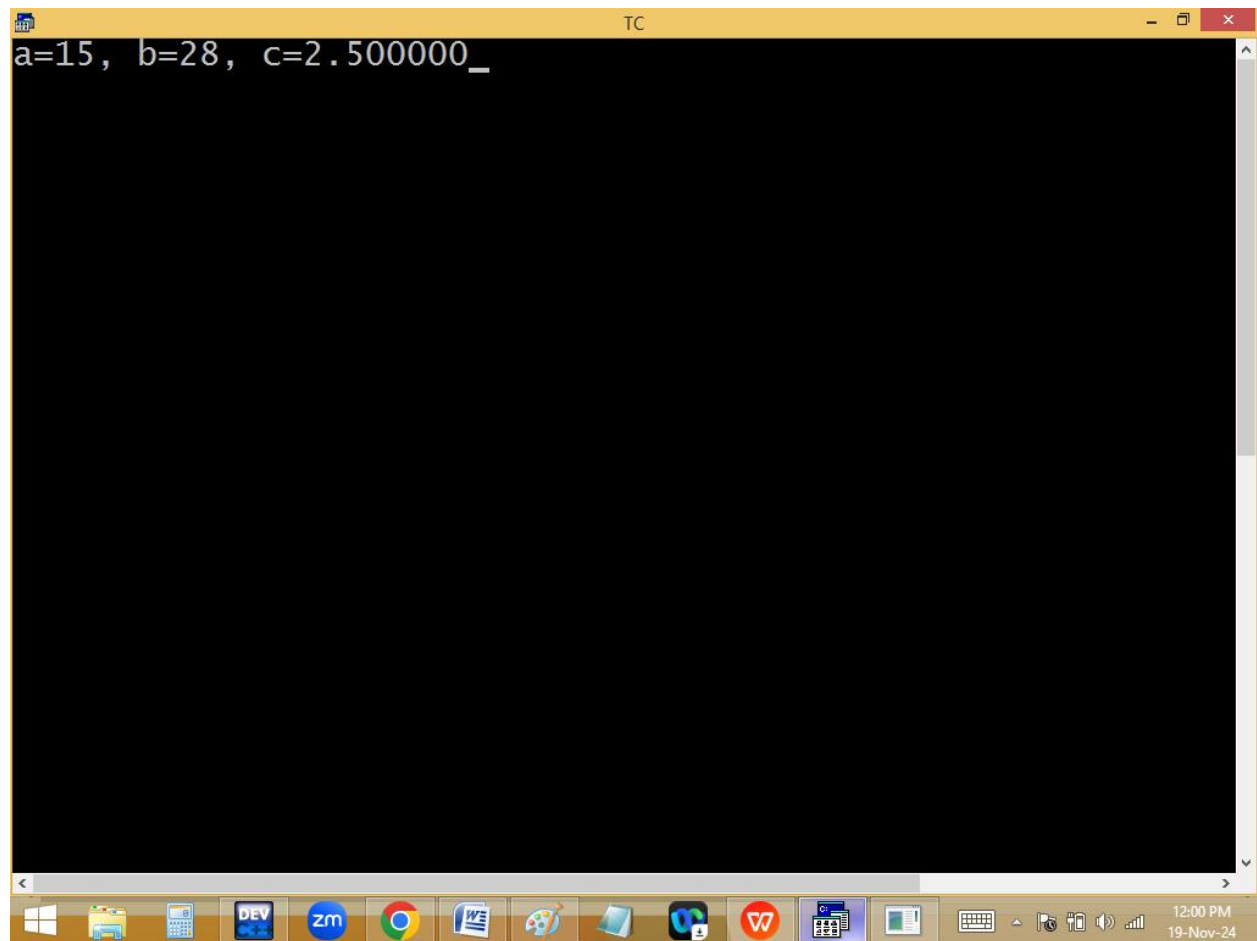
c/=2; i.e. $c=c/2 \rightarrow c=5/2=2.500000$

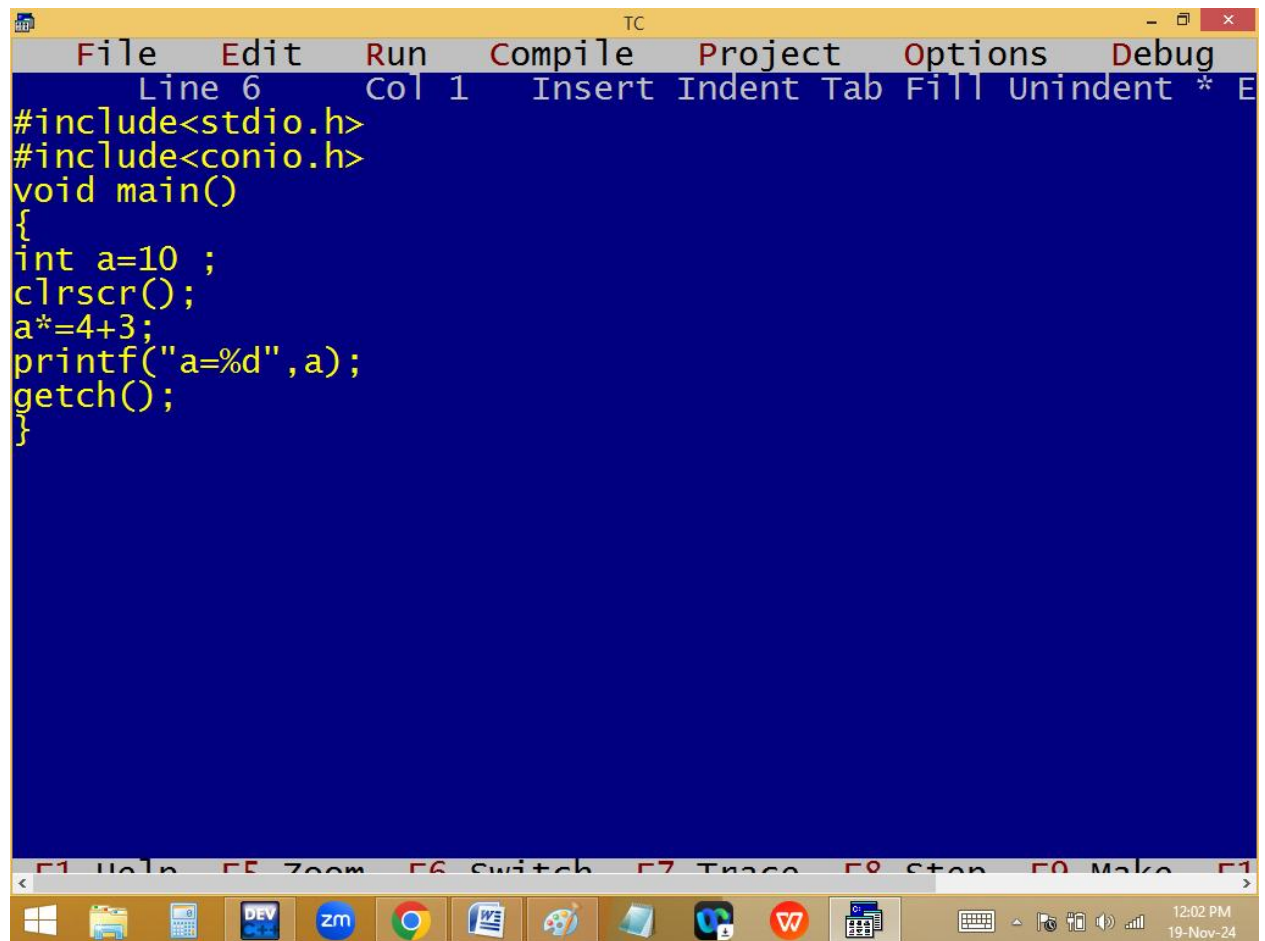


```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, b=7;
float c=5;
clrscr();
a+=5;
b*=4;
c/=2;
printf("a=%d, b=%d, c=%f",a,b,c);
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:00 PM
19-Nov-24

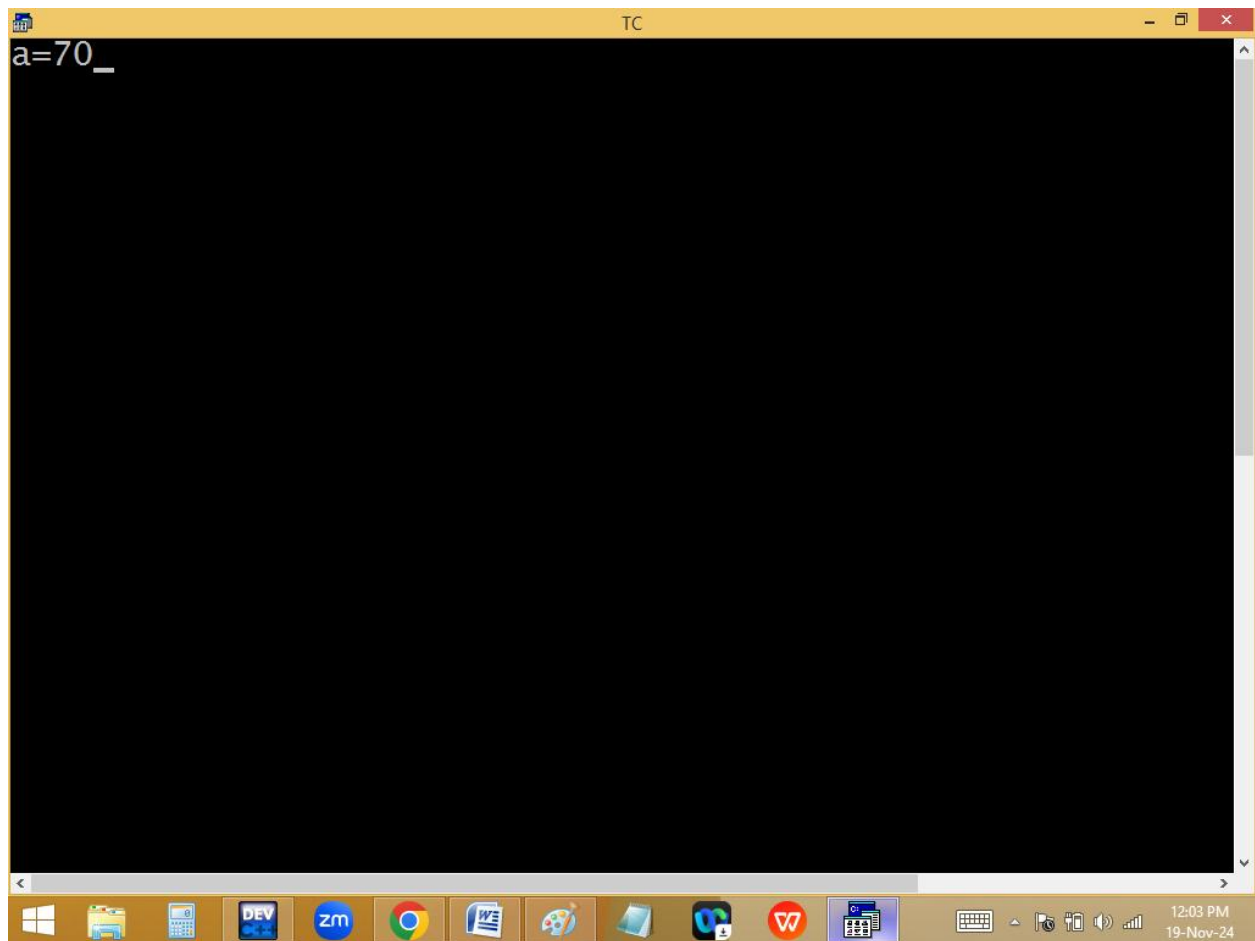




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 6", "Col 1", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background and contains the following C code:

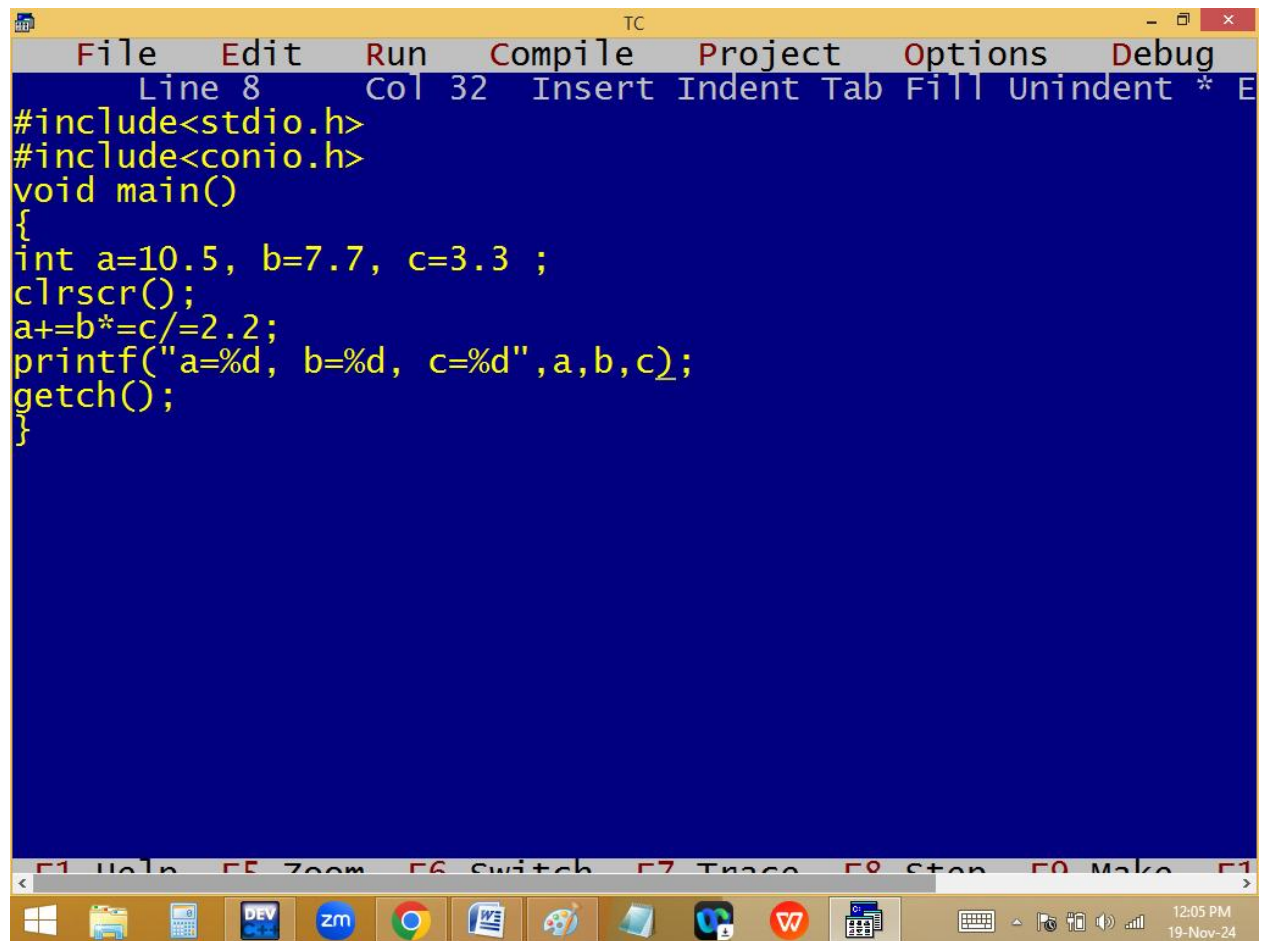
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10 ;
clrscr();
a*=4+3;
printf("a=%d",a);
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and other applications. The system clock in the bottom right corner shows "12:02 PM" and "19-Nov-24".



a=10

a*=4+3; ==> a*=7 ==> a=a*7 ==> a=10*7=70



```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 32 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10.5, b=7.7, c=3.3 ;
clrscr();
a+=b*c/=2.2;
printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:05 PM 19-Nov-24

```
TC
a=17, b=7, c=1_
```

int a=10, b=7, c=3;

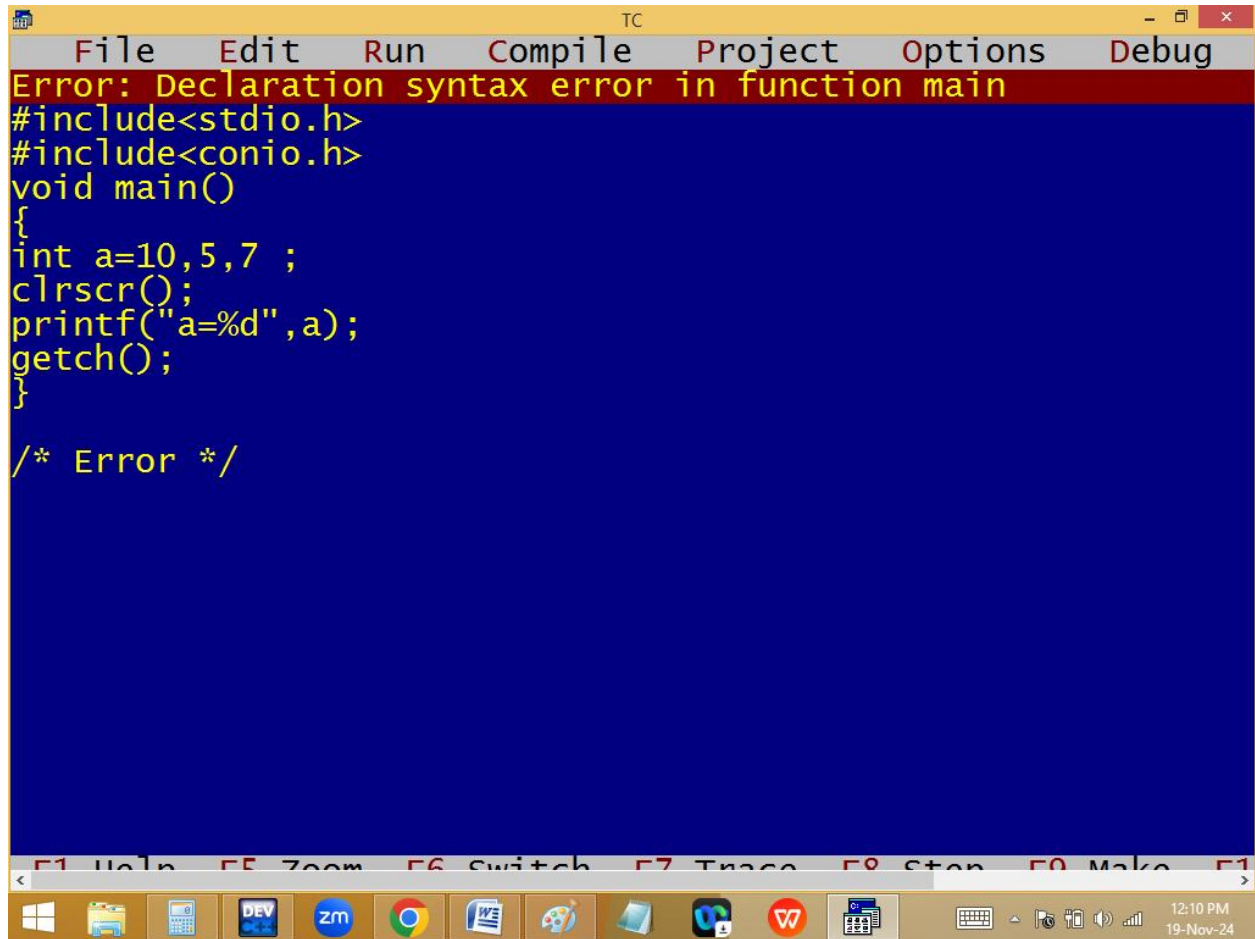
a+=b*=c/=2.2;

c=c/2.2 ==> 3/2.2=1

b*=c ==> b=b*c ==> b=7*1=7

a+=b ==> a=a+b ==> a=10+7=17

() and , separators:



The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. A red error message banner at the top reads "Error: Declaration syntax error in function main". The code editor contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10,5,7 ;
clrscr();
printf("a=%d",a);
getch();
}

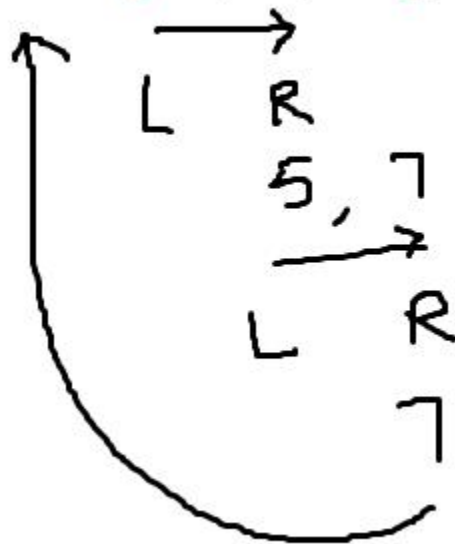
/* Error */
```

Below the code editor, there is a toolbar with icons for various functions like Help, Zoom, Switch, Trace, Stop, and Make. The Windows taskbar at the bottom shows the system clock as 12:10 PM on 19-Nov-24.


```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 7 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=(10,5,7) ;
clrscr();
printf("a=%d",a);
getch();
}

/* a=7 */
```

int a = (10, 5, 7);



```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 32 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* declaration */
a=10,5,7 ; /* initialization */_
clrscr();
printf("a=%d",a);
getch();
}

/* a=10 */
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```


a = 10, 5, 7;

```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 9 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* declaration */
a=10,(5),7 ; /* initialization */
clrscr();
printf("a=%d",a);
getch();_
}

/* a=10 */
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

a = 10, (5), 7;
↓
a = 10, 5, 7;
↖

```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 11 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* declaration */
a=10,(5,7) ; /* initialization */
clrscr();
printf("a=%d",a);
getch();
}

/* a=10 */
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

a = 10, (5, 7);

→

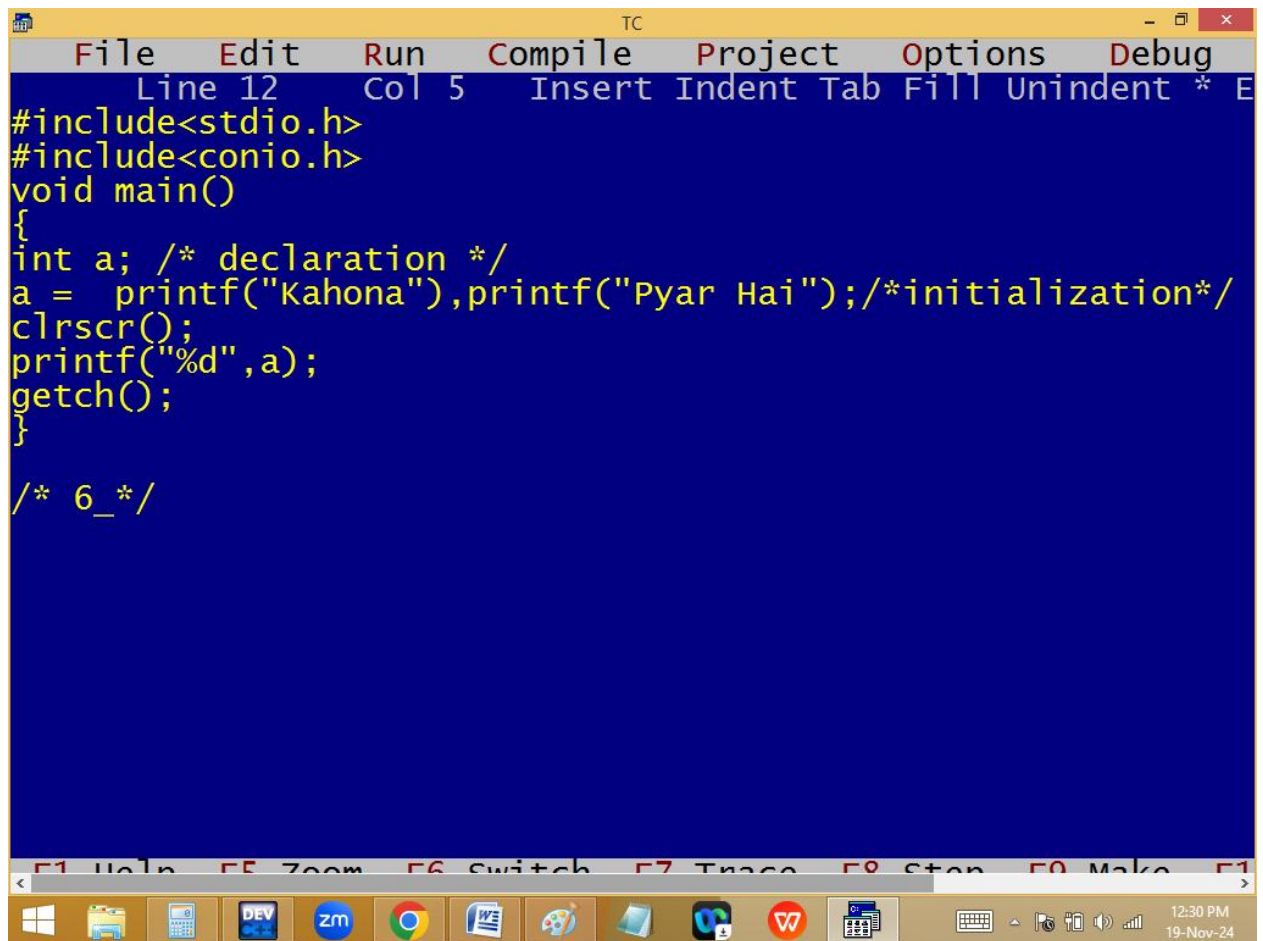
a = 10, 7

↖

```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 7 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* declaration */
a=(10,3),(5,7),9 ; /* initialization */
clrscr();
printf("a=%d",a);
getch();
}

/* a=3_*/
```

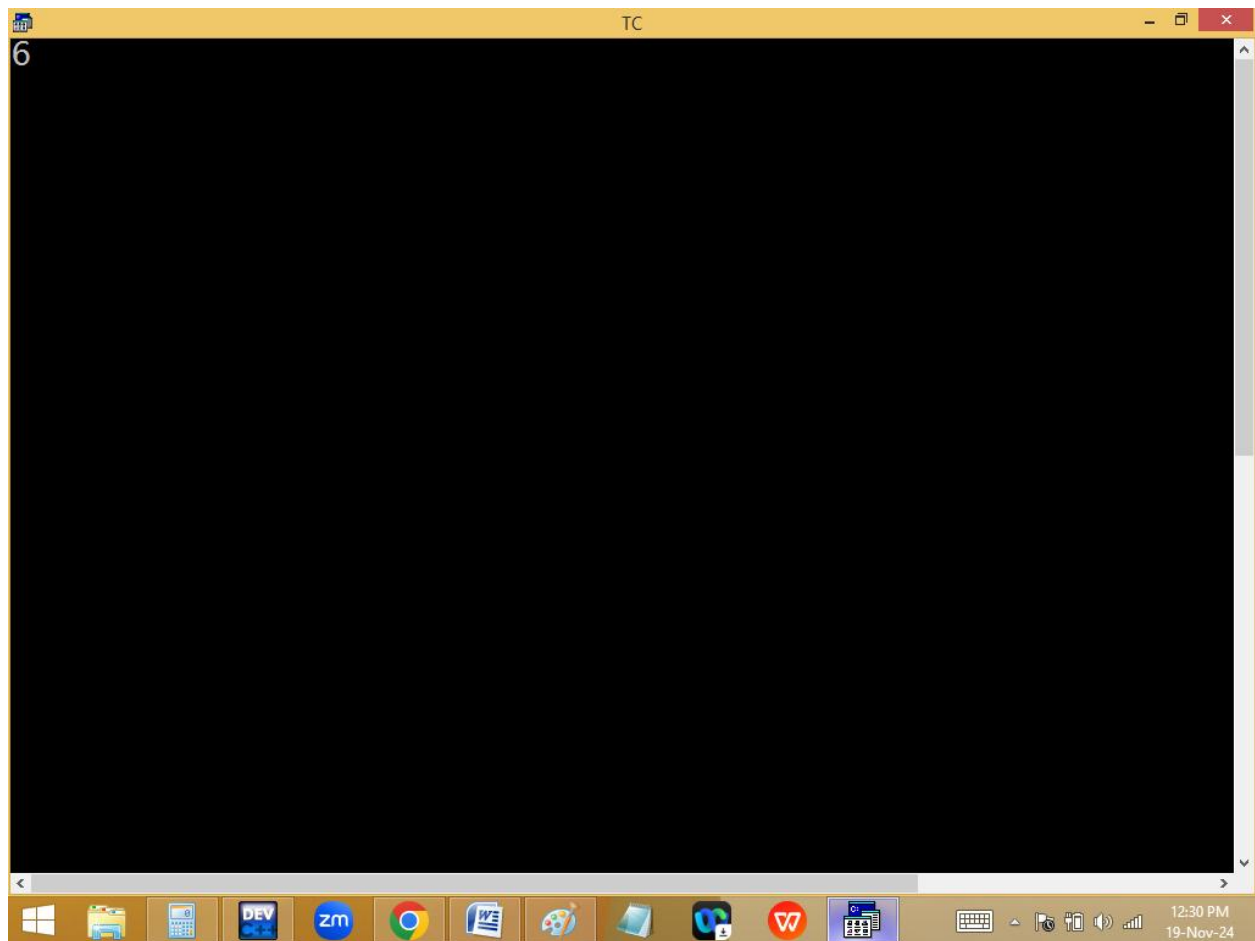
a = (10, 3), (5, 7), 9;
→ →
a = 3 , 7 , 9
↖



```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 5 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* declaration */
a = printf("Kahona"),printf("Pyar Hai");/*initialization*/
clrscr();
printf("%d",a);
getch();
}

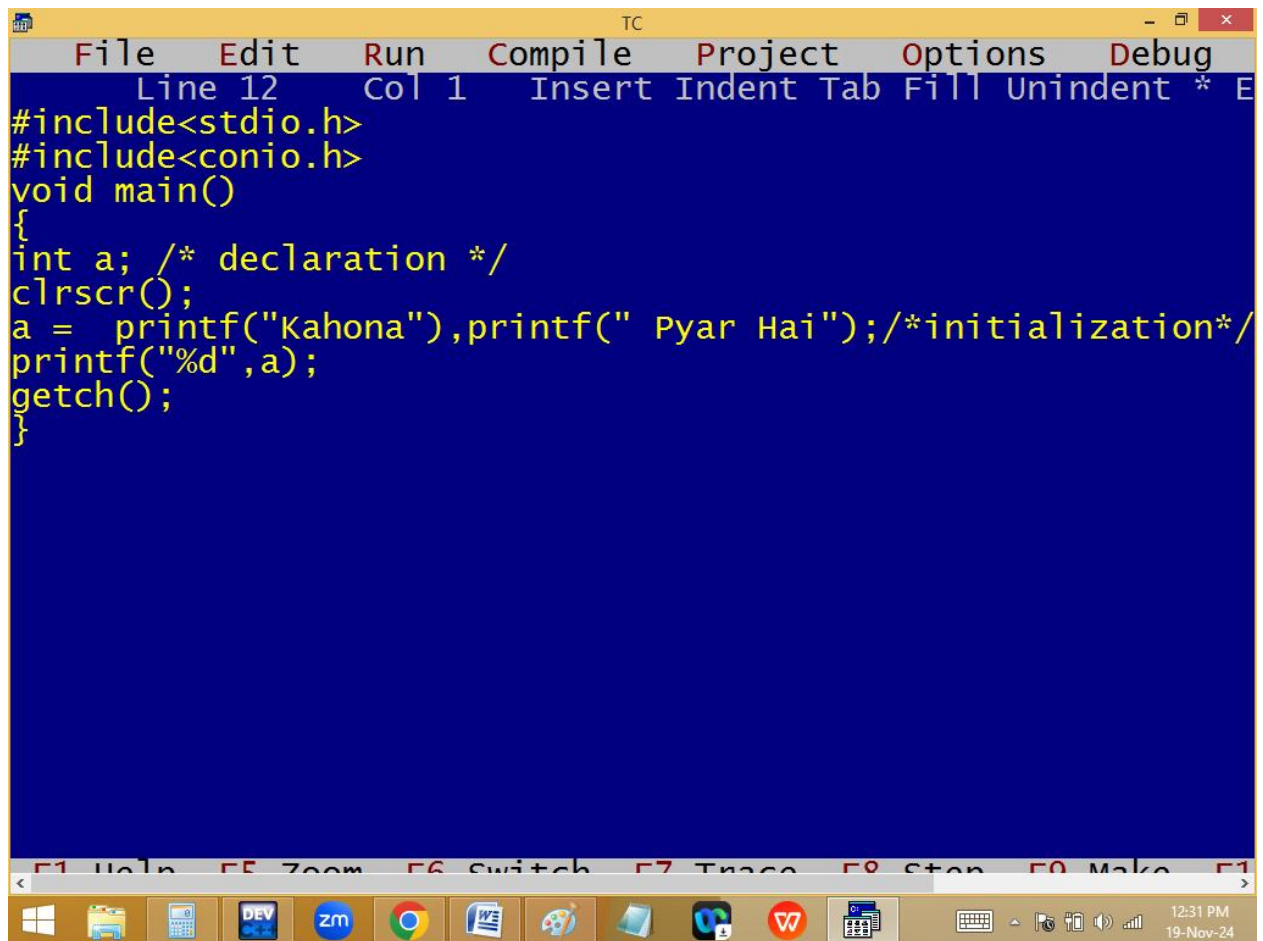
/* 6_*/

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```



✓
a = p("Kahona"), p("Pyar Hai");

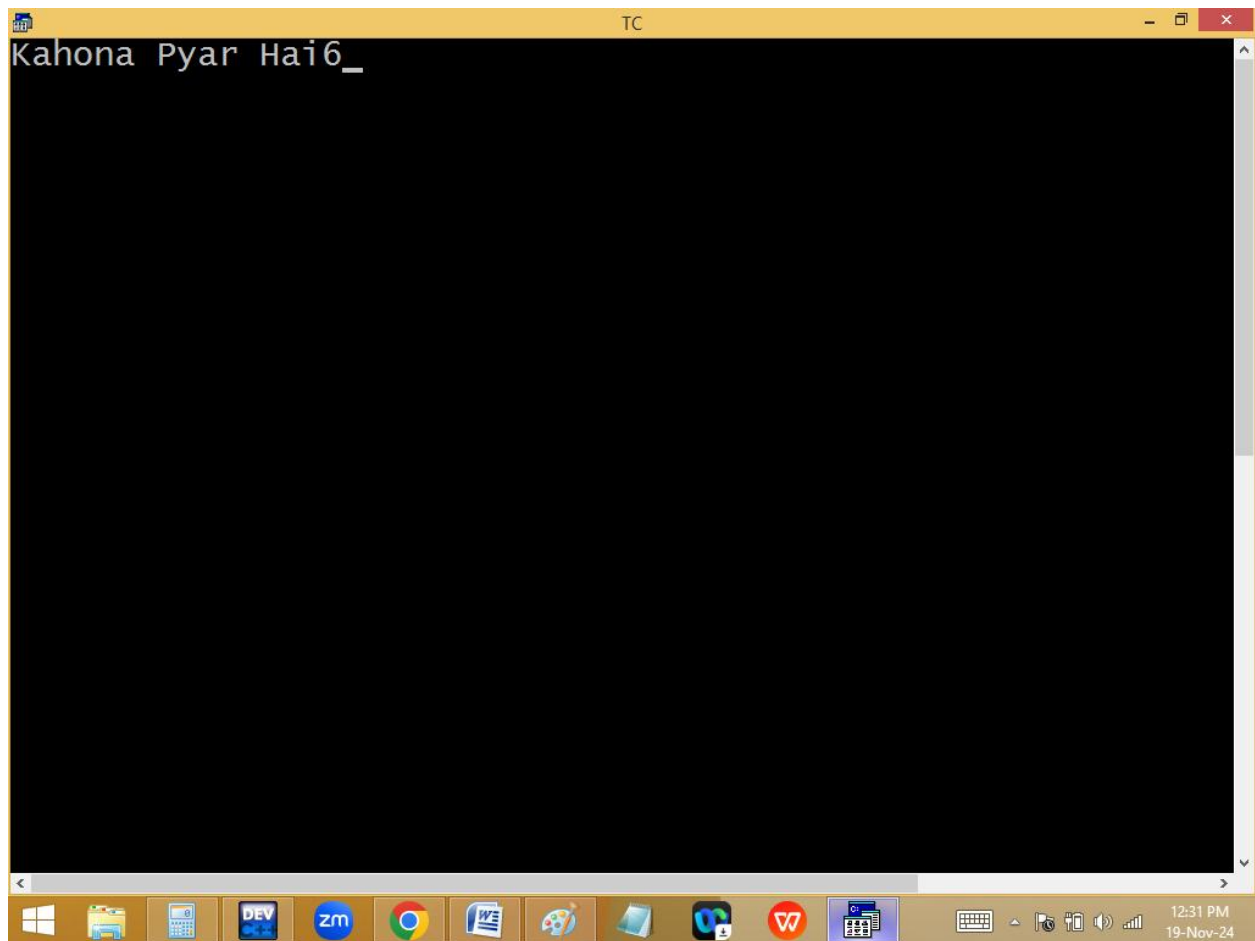
a = 6 , 8
└──┘



```
TC
File Edit Run Compile Project Options Debug
Line 12 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* declaration */
clrscr();
a = printf("Kahona"),printf(" Pyar Hai");/*initialization*/
printf("%d",a);
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:31 PM
19-Nov-24



Address operators:

1. & - address of the variable
2. * - pointer – address of another variable

```
Int a=10.5;
```